A 31-year (1990-2020) global gridded population dataset generated by cluster analysis and statistical learning

Luling Liu[1,2], Xin Cao[1,2], Shijie Li[1], Na Jie[1]

Introduction

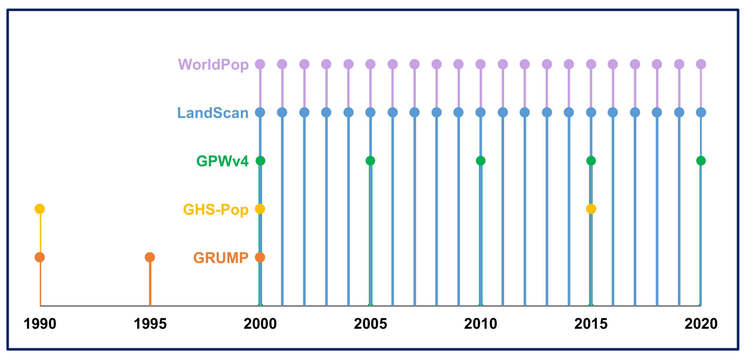

Fig.1 Gridded population data time frequency

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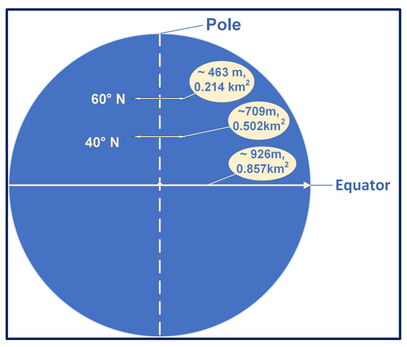

Fig.2 Distortions of WGS-84 coordinate system

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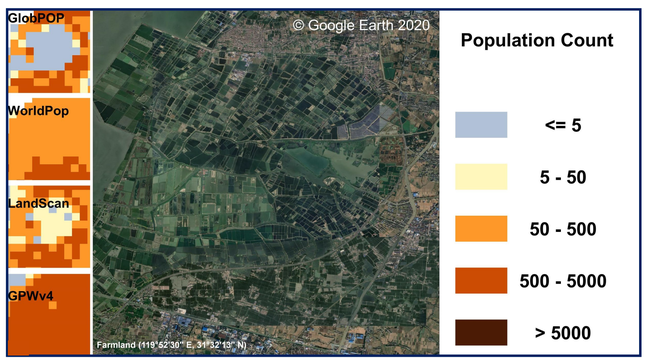


Fig.3 Overestimation in uninhabited areas

Objective

To integrate five sets of multi-source demographic data and leverage the strengths of each data, and produce a new population product (**GlobPOP**) suitable for long time-series analysis.

Result 1

The first finding reveals that GlobPOP has a high level of accuracy in predicting country-level population estimates comparing with the World Population Prospects 2022.

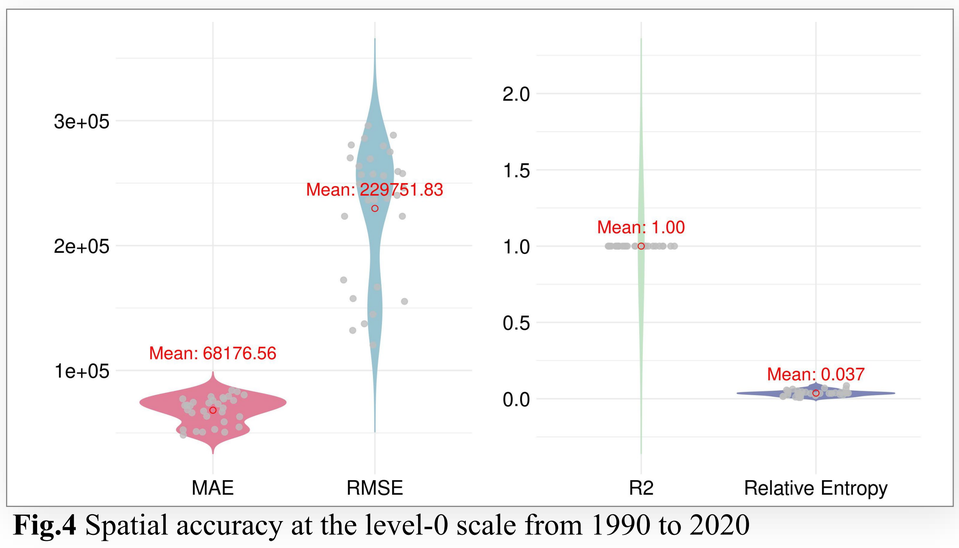


Fig.4 Spatial accuracy at the level-0 scale from 1990 to 2020

Result 2

The figure compares population time-series curves for eight countries using the GlobPOP dataset and five others, highlighting GlobPOP's superior temporal accuracy.

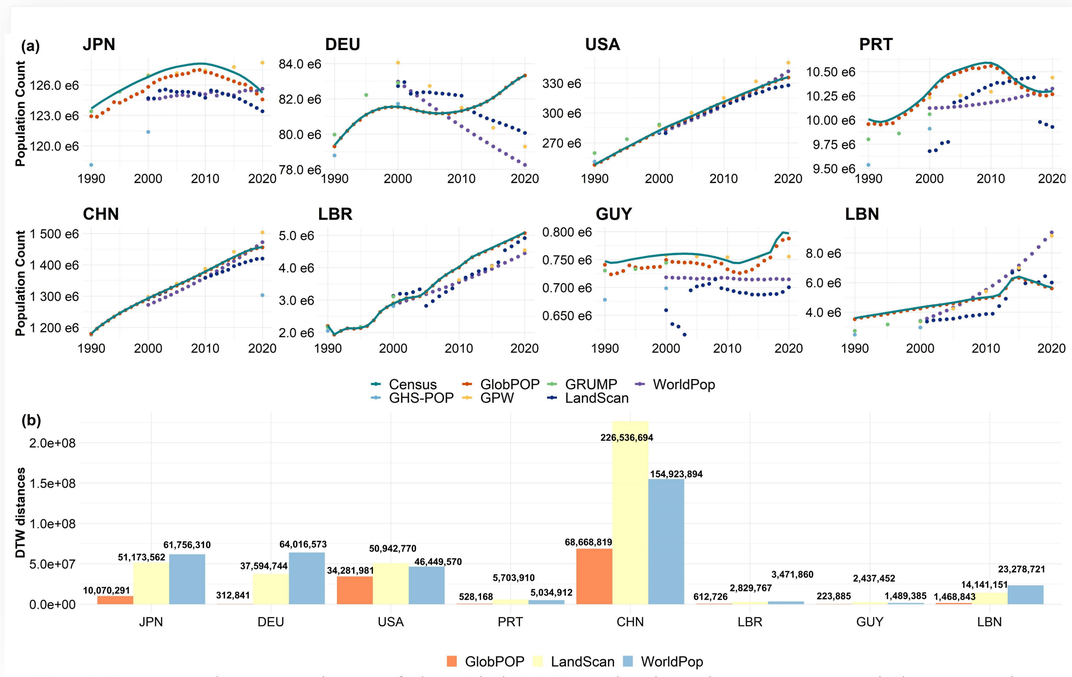


Fig.5 Temporal comparison of the GlobPOP and other datasets over eight countries

Result 3

The pixels with population higher than 5,000 had increased significantly in India, China, western Europe, the eastern and southern United States, and South Sahara Africa since 1990.

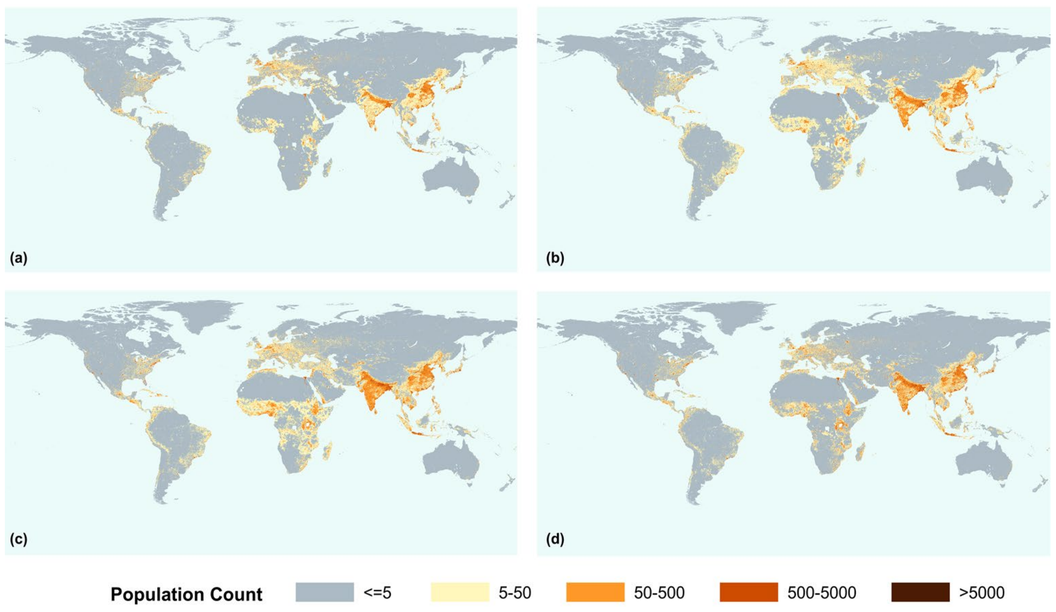


Fig.6 The global gridded population distribution(1990-2020)

Materials and methods

This study introduces a novel data fusion framework to create a continuous global gridded population dataset, GlobPOP, spanning from 1990 to 2020.

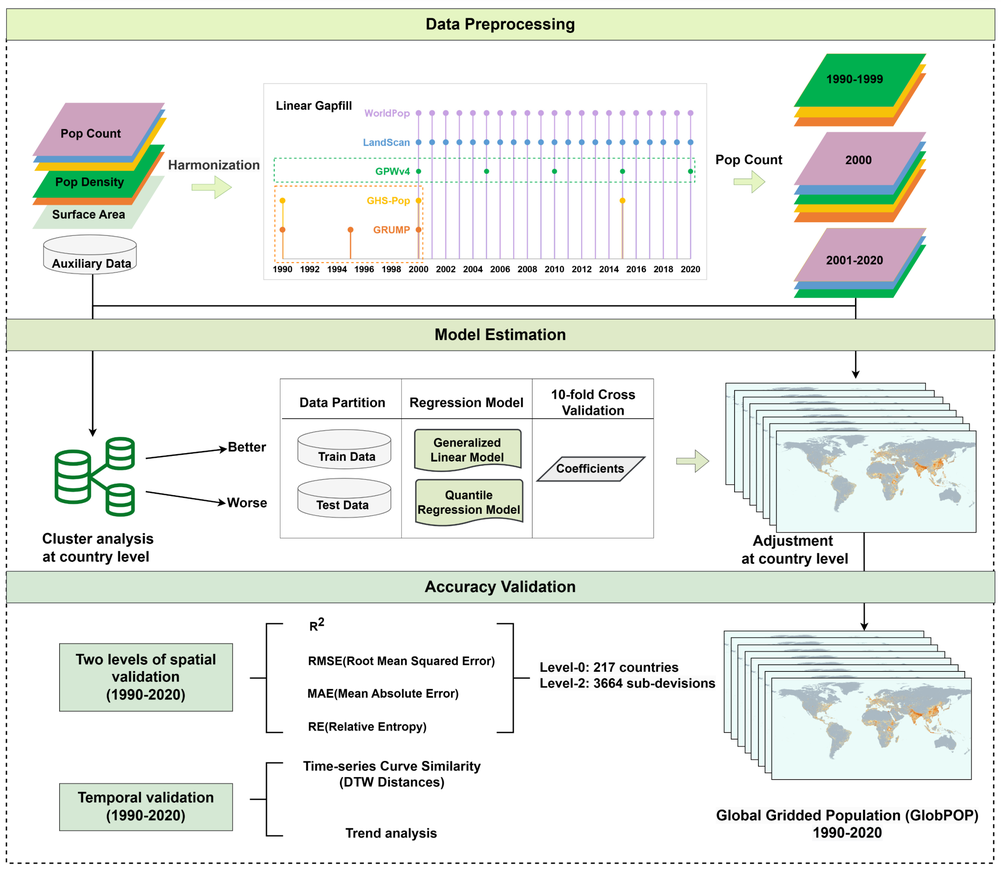
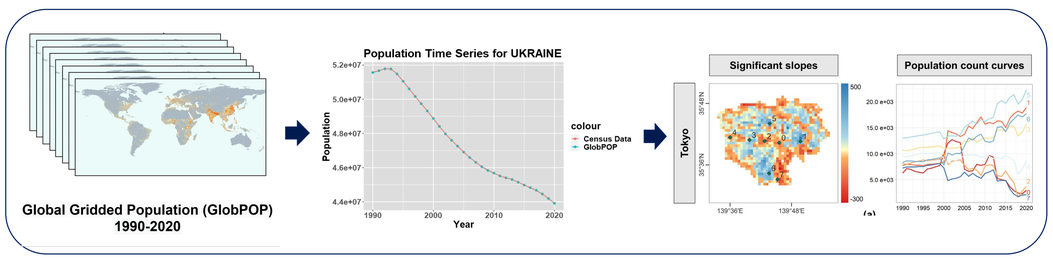


Fig.7 Workflow of the estimation and validation of the global gridded population (GlobPOP)

Conclusion



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Affiliations

1. State Key Laboratory of Remote Sensing Science, Faculty of Geographical Science, Beijing Normal University
2. Beijing Engineering Research Center for Global Land Remote Sensing Products, Faculty of Geographical Science, Beijing Normal University

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Contact:
lulingliu@mail.bnu.edu.cn